



Schneider Electric

Macropad

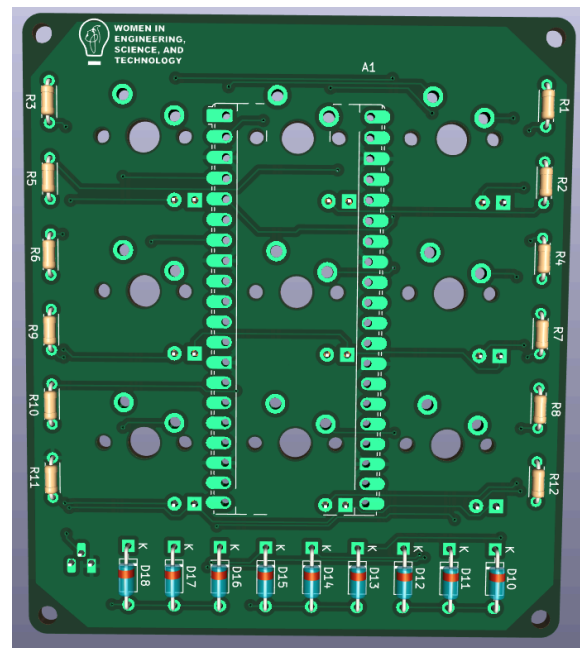
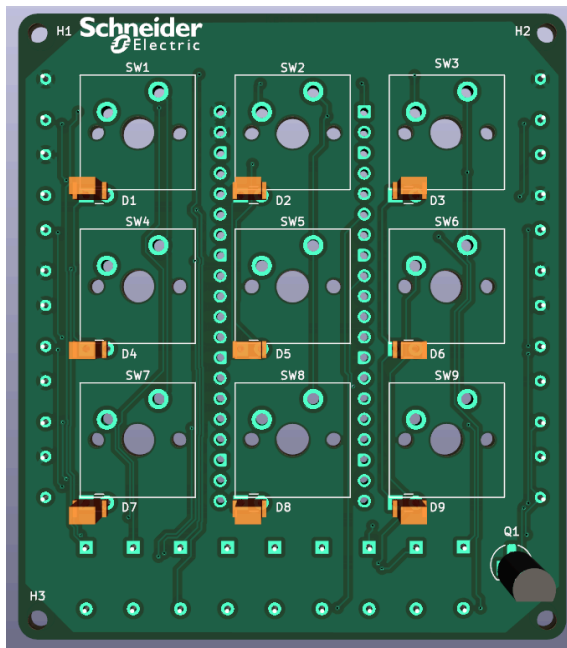
WEST Project

Victoria, BC, Canada

Layout Design

Asil Helali

Summer 2025



Life Is On

Schneider
Electric

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1.0 Board Setup

This section of the document is going to walk you through how to setup the KiCAD tool to start your layout design.

1.1 Annotate the Schematic

Before starting the layout design, we must ensure that all components are annotated on the schematic. This is done to allow the tool to import successfully without any errors from schematic to layout. It provides proper synchronization and its good practice for organization.

Open the Schematic Editor → Tools → Annotate the Schematic → Reset Existing Annotations → Annotate.

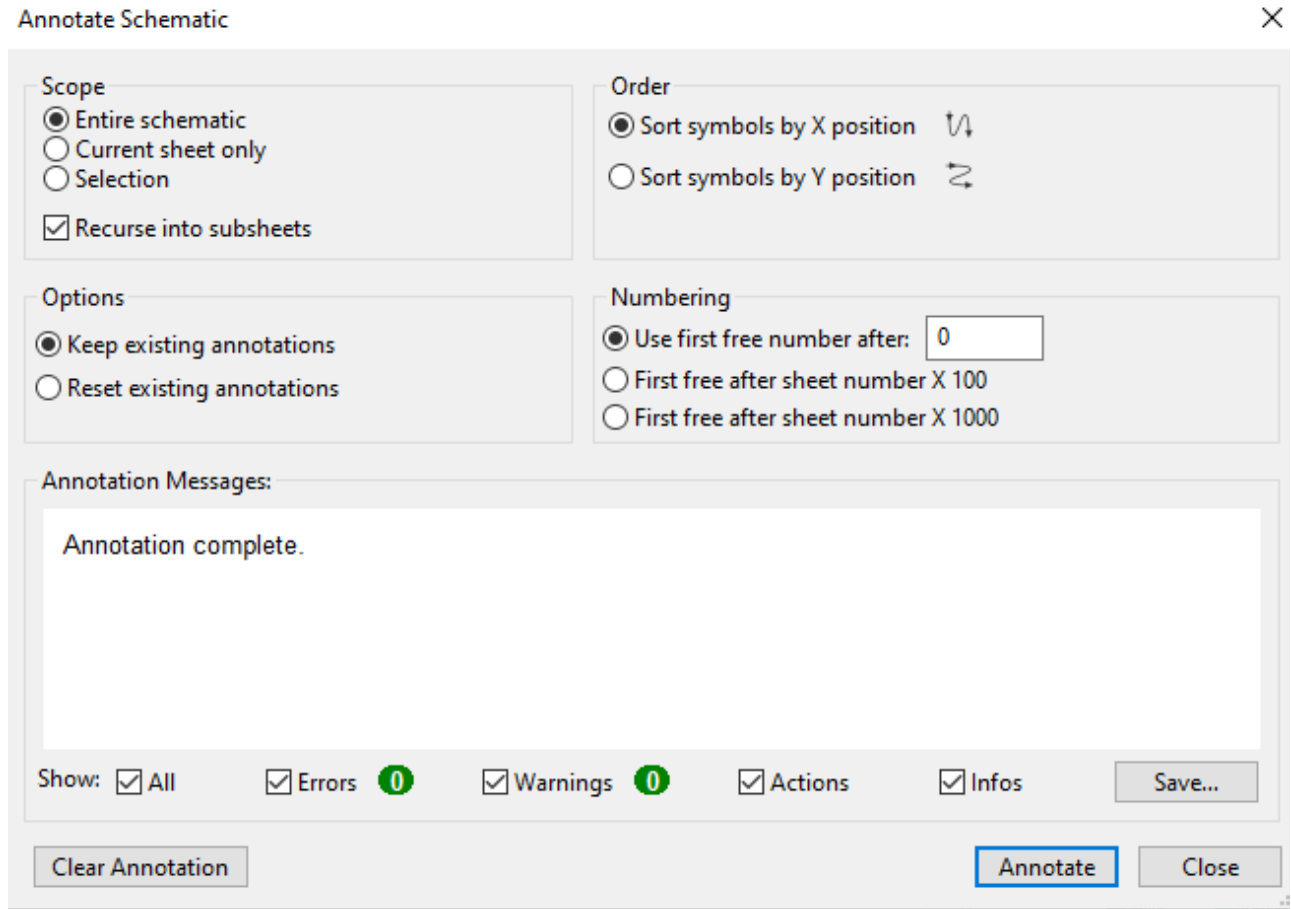


Figure 1: Annotate Schematic

1.2 Footprint Library Sync

KiCad has a basic footprint library that is automatically installed, however for more specific footprints (eg. switches) extra footprint plugins need to be downloaded. In the main KiCAD window go to **Tools → Plugin and Content Manager**, then install the **Keyswitch Kicad Library**.

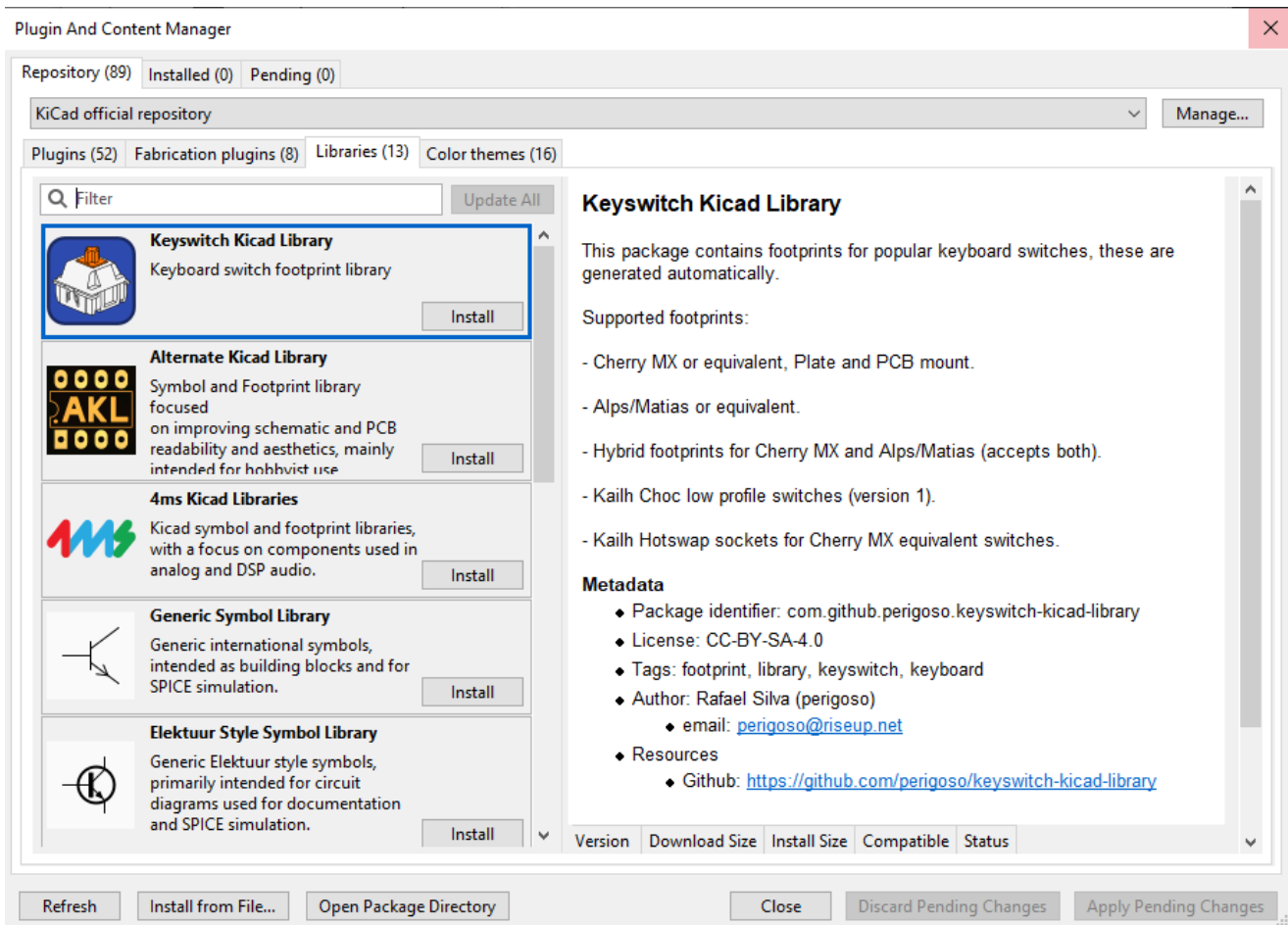


Figure 2: Downloading Footprint Library

Note: This does not need to be done for this project, all parts were already included in advance.

1.3 PCB Stackup

The stackup of a PCB refers to the specific layering of conductive (copper) and insulating (dielectric) materials that make up a printed circuit board. The stackup defines how many layers the PCB has, the thickness of each layer, the material and their order. For this project, we will be making a 2 copper layer board. To look at the stack up **go to File → Board Setup→ Physical Stackup**. From there you can choose the amount of layers you want your board to be.

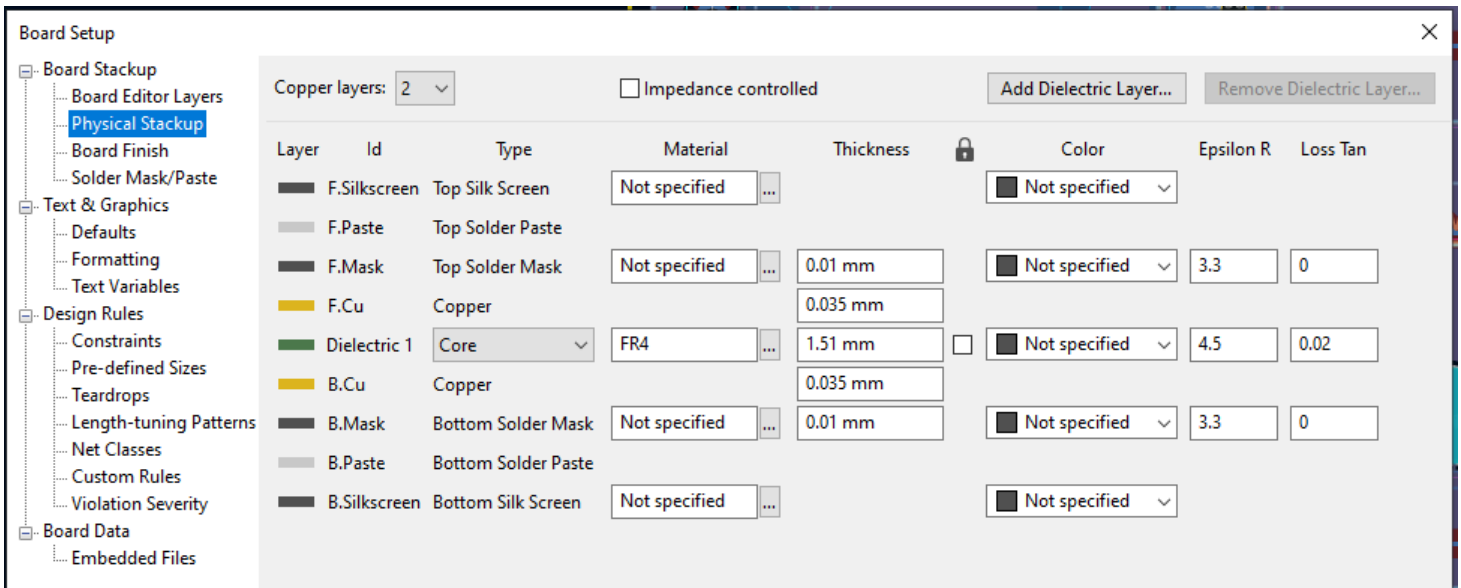


Figure 3: Board Stackup

In the stackup, there is a total of 9 layers for a 2 copper layer board:

- **Silkscreen:** top most surface layer, non conductive, it's the “writing” on the board, which provides essential information like component identifiers, polarity, test points, specific instructions and logos.
- **Solder Paste:** This layer defines the areas on the board where solder paste is applied on surface mount components.
- **Solder Mask:** This layer is a protective polymer coating that insulates copper traces and prevents solder bridges between closely placed pads by covering all areas except for the solderable pads and holes.
- **Front Copper:** This is a copper plane on the board, it's used to form conductive pathways, pads and lands for electrical connections to components. It provides electrical flow, signal transmission, power distribution and heat dissipation.

- **Dielectric Core:** This is a solid non-conductive layer that provides mechanical stability and electrical insulations between copper layers. The copper layer is laminated on one (or both sides) of this core. It prevents electrical shorts between conductive layers.
- **Back Copper:** Same as Front Copper layer, on the other side of the dielectric core.
- **Prepreg:** Shown in Figure 3, this layer is made up of fiberglass impregnated with partially cured resin. It acts as an adhesive to bond together the copper cores and other layers. It provides mechanical strength, electric isolation.

Note: F and B refer to Front and Back.

In the Board Setup window, you can specify the material for each layer as well as the thickness. The core material that we will be using for this project is FR4 (flame retardant 4) which is a type of epoxy resin that self-extinguishes if ignited which enhances safety for the PCB.

The layer thickness is standardized. The most common PCB thickness is 1.57mm (62mils). When choosing the stackup thickness, keep in mind the current, environmental factors and space constraints, impedance and cost constraints.

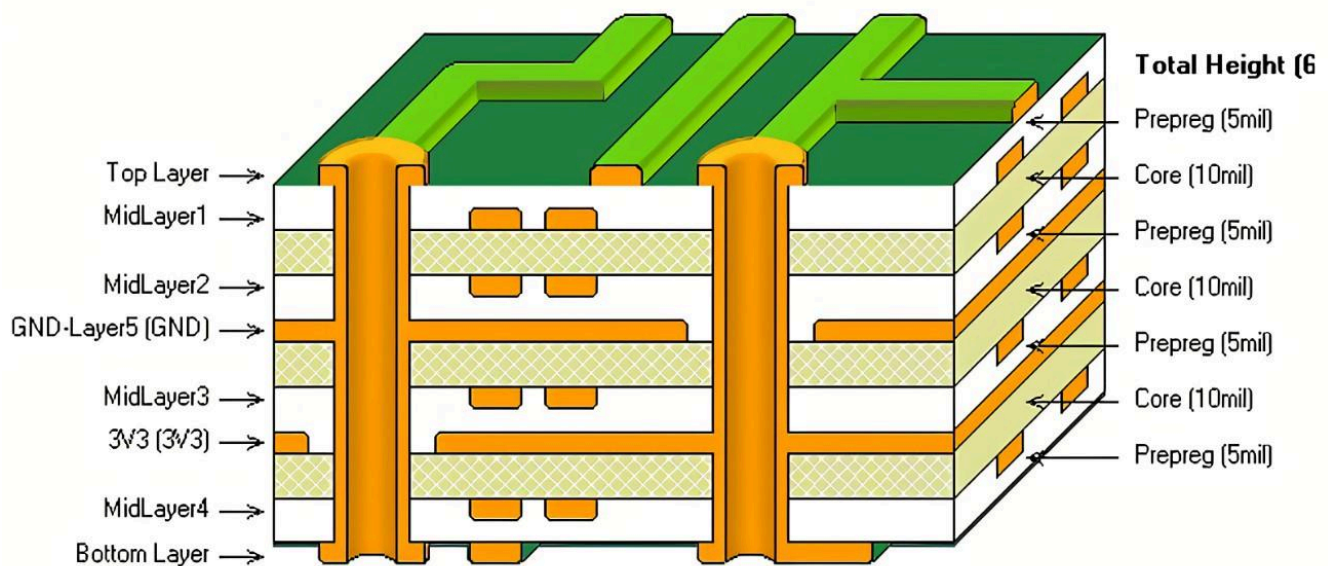


Figure 4: Exploded PCB Stackup View

1.4 Constraints Manager

The first thing you have to do is set up your board constraints. These are the manufacturing limitations that make sure the boards are working as intended. They prevent issues like soldering defects, assembly problems and shorts.

These constraints are taken directly from the JLC website which is the manufacturer that we will be using for this project. [PCB Manufacturing & Assembly Capabilities - JLCPCB](#)

To edit your constraints **go to File → Board Setup → Constraints**. Copy the values from the figure below.

Board Setup

- Board Stackup
 - Board Editor Layers
 - Physical Stackup
 - Board Finish
 - Solder Mask/Paste
- Text & Graphics
 - Defaults
 - Formatting
 - Text Variables
- Design Rules
 - Constraints**
 - Pre-defined Sizes
 - Teardrops
 - Length-tuning Patterns
 - Net Classes
 - Custom Rules
 - Violation Severity
- Board Data
 - Embedded Files

Copper

	Minimum clearance:	0.15	mm
	Minimum track width:	0.15	mm
	Minimum connection width:	0.15	mm
	Minimum annular width:	0.08	mm
	Minimum via diameter:	0.26	mm
	Copper to hole clearance:	0.29	mm
	Copper to edge clearance:	0.5	mm

Holes

	Minimum through hole:	0.16	mm
	Hole to hole clearance:	0.21	mm

uVias

	Minimum uVia diameter:	0.25	mm
	Minimum uVia hole:	0.15	mm

Silkscreen

Arc/Circle Approximated by Segments
Maximum allowed deviation: 0.005 mm
Note: zone filling can be slow when < 0.005 mm.

Zone Fill Strategy
 ☒ Allow fillets/chamfers outside zone outline
 Minimum thermal relief spoke count: 2

Length Tuning
☒ Include stackup height in track length calculations

Import Settings from Another Board... OK C

Figure 5: Constraint Values

1.5 Uploading the Mechanical Outline

Mechanical outlines are provided to give guidance and show the boundaries of the enclosure. They provide insight on the height restrictions, location of switches and where the screw holes are. They come in the form of a .DXF file, which will be provided.

To import the outline **go to File → Import → Graphics.**

When the “**Import Vector Graphics File**” window appears, locate and **upload the .DXF file**. Make sure to change the “**DXF default units**” to **millimeters** and **double check the layer selected is “User.Drawings”**. Click “OK” and place the outline within the sheet.

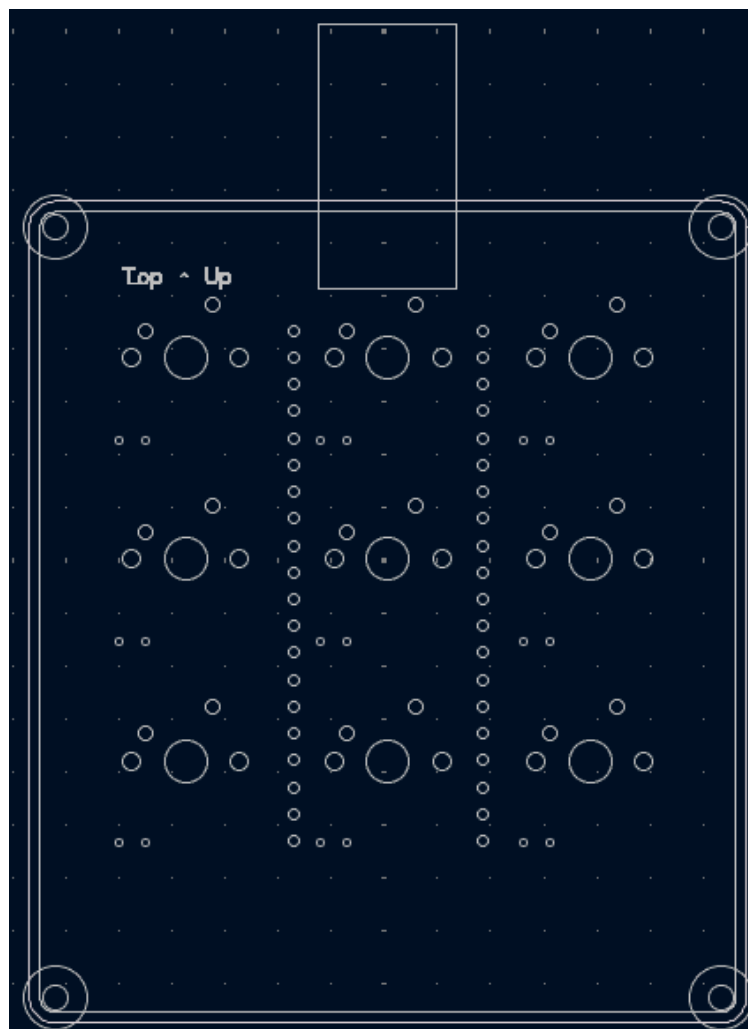


Figure 6: Mechanical Outline

1.6 Schematic to PCB Sync

Once the libraries are all installed, we can now upload all components from the schematic to the PCB. In the PCB Editor window, go to **Tools** → **Update PCB from Schematic**. Make sure there are no Errors or Warnings and then click **“Update PCB”** and place all the components to the side.

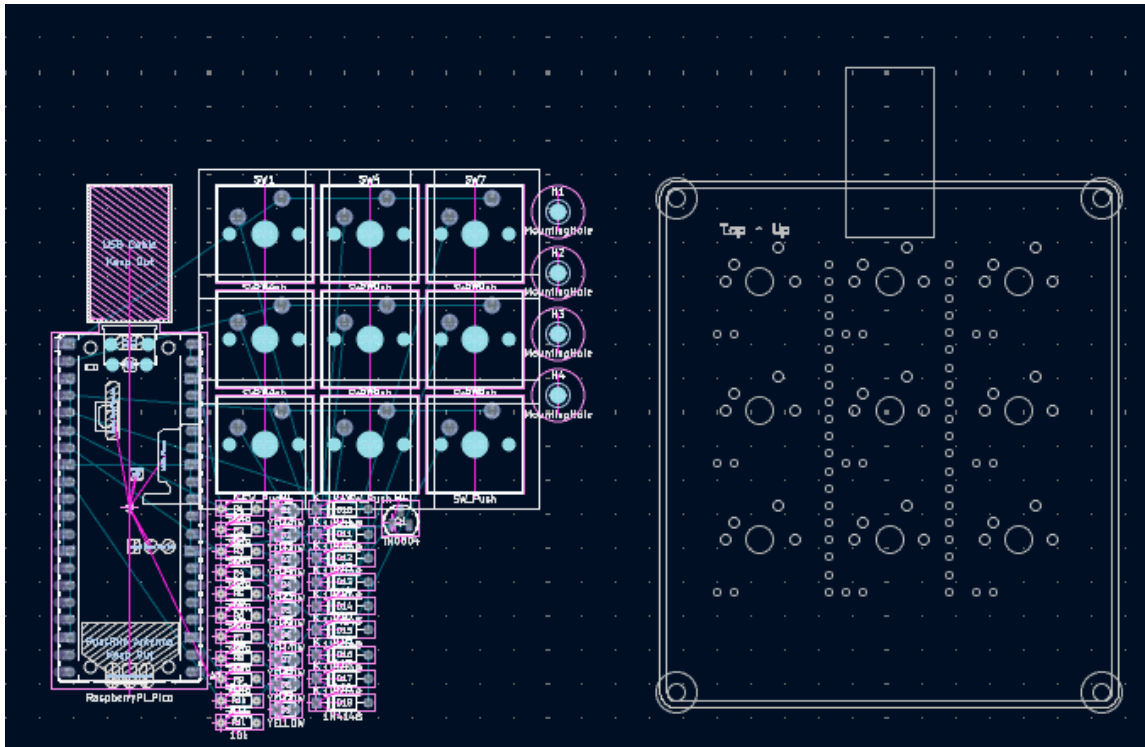


Figure 7: Footprints Placed

2.0 Board Design

This section of the document is going to walk you through how to start and complete your layout design.

2.1 Creating the Board Outline

In this section we will be creating the board outline. **Right click on the outer mechanical drawing boundary and go to properties.** In the **“Segment Properties”** window, change the layer to **“Edge.Cuts”**. Each segment on the outline will need to be done individually.

Note: If you are unable to select each segment individually, you might need to **“ungroup”** the section by right clicking → **Grouping** → **Ungroup Items**.

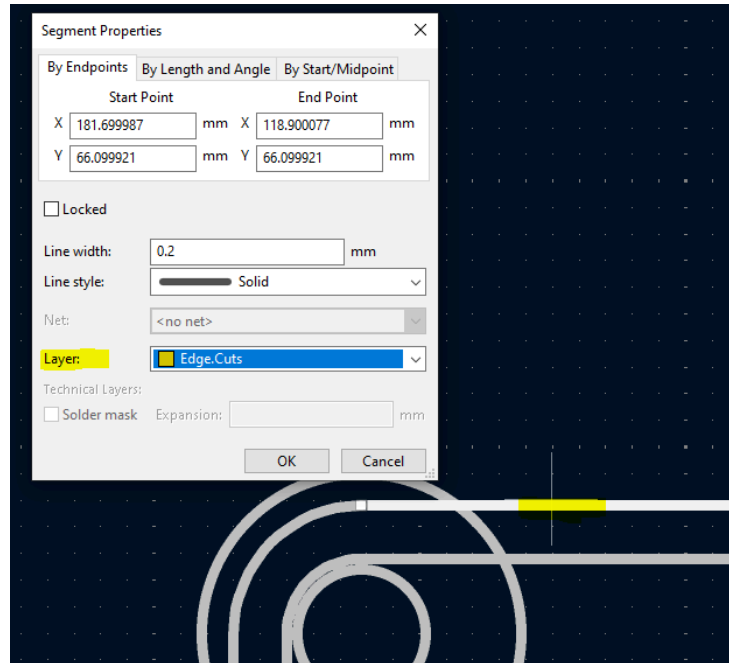


Figure 8: Segment Properties Example

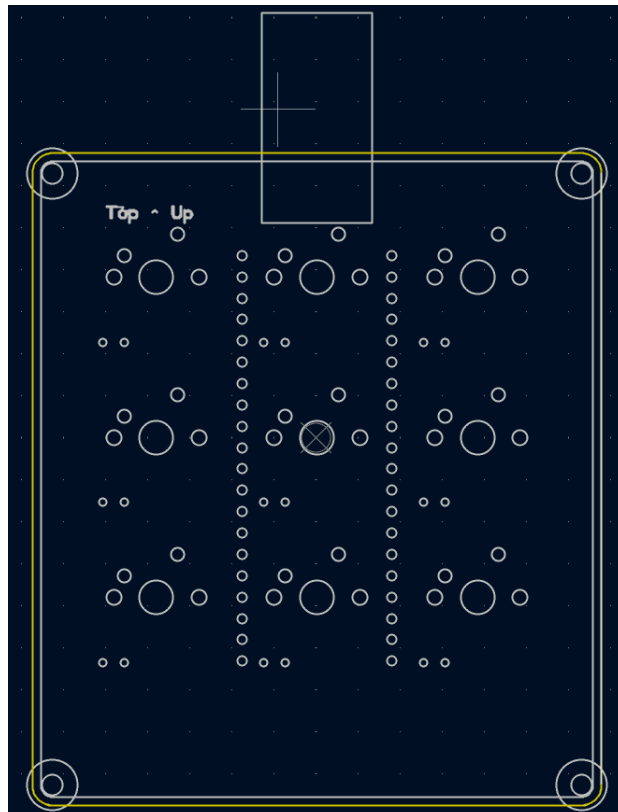


Figure 9: Edge.Cuts Outline Completed

Once the board outline is completed, make sure to **lock both the “Edge.Cuts” and “User.Drawings” layers by right clicking → Locking → Lock**. This ensures that the outlines do not move. If you want to edit / unlock a locked item, make sure the “Locked Items” button is selected in the “Selection Filter” tab on the bottom right section.

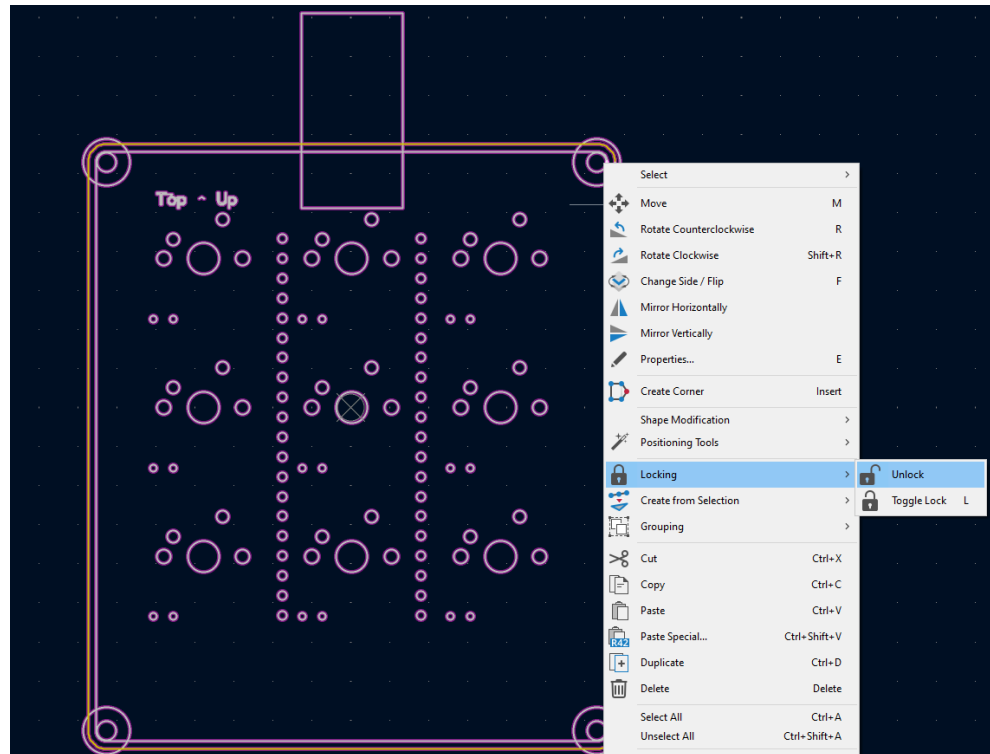


Figure 10: Locked Layers Example

2.2 Creating Copper Planes and Assigning Nets

This is a 2- layer board so we will need to create 2 copper planes. The top plane will be the power plane, which is +5V and the bottom plane will be ground. Planes are mainly used to evenly distribute voltage across the PCB and to provide a stable connection to all components.

The copper plane provides an area for electrical signals to flow between components. This is done by forming traces and pads and connections on the copper plane.

All conductive planes will be at least 0.5mm less than the board outline. This is done to prevent any accidental copper exposure that could cause shorts.

The inner mechanical outline is about 0.8mm from the board outline (this can be checked using the measuring tool) and it is where we will place the copper planes.

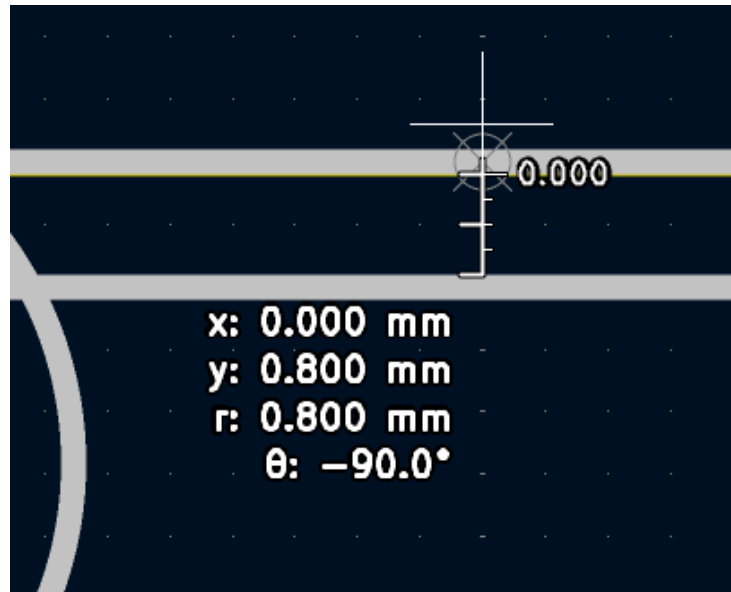


Figure 11: Board Outline Measurement

To create a plane go to **Place** → **Draw Filled Zones**. Once you select a corner to start, the “Copper Zone Properties” window will appear. Make sure to select the Layer and the Net desired for that layer (+5V for F.Cu and GND for B.Cu), only one layer is done at a time.

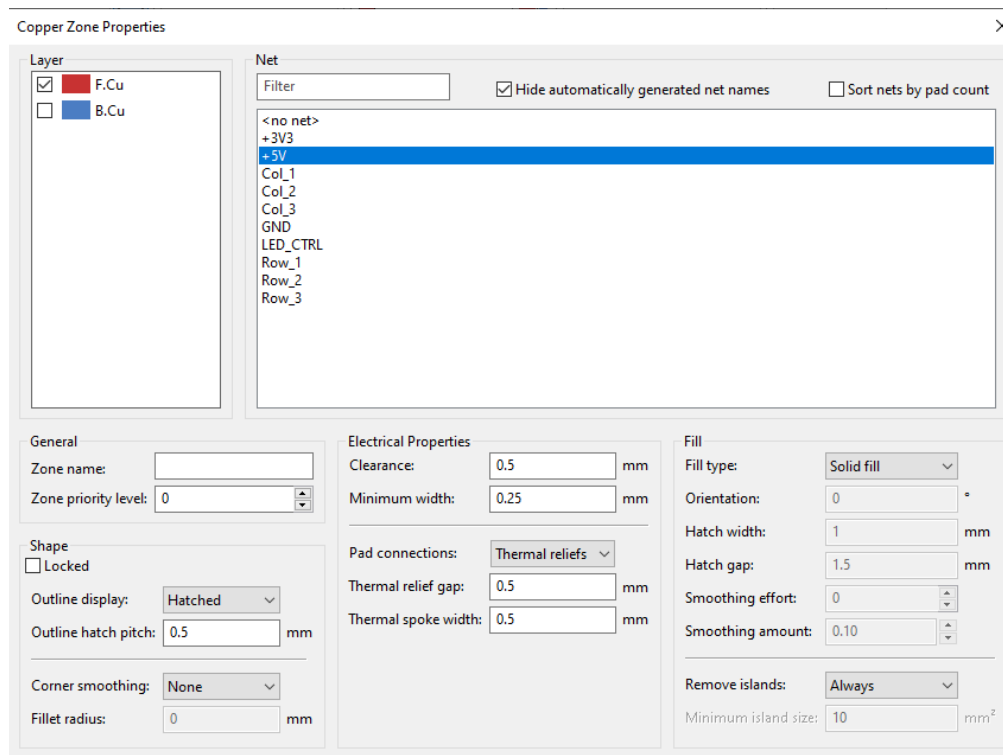


Figure 12: Copper Zone Properties

Note: To prevent any conductive exposure to the drill holes, make sure to not include the drill holes within the copper section. See figure below for an example.

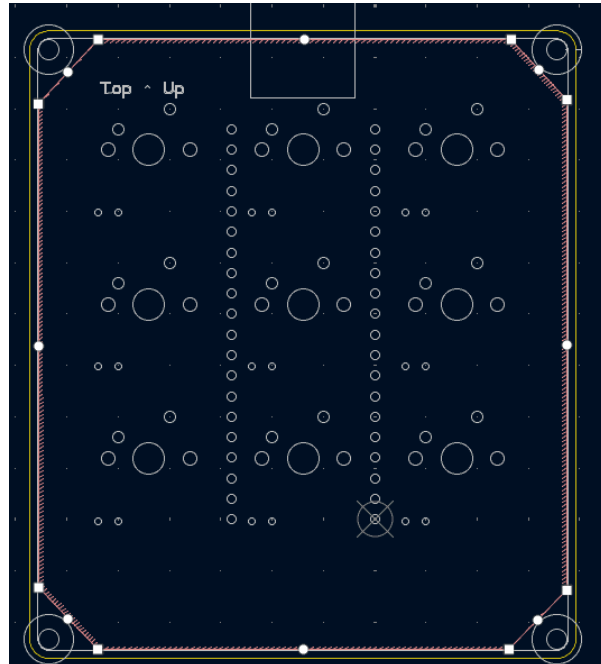


Figure 13: Copper Zone

Once the outline is complete, you can fill the copper zone by going to **Edit**→ **Fill All Zones** or by pressing “B”.

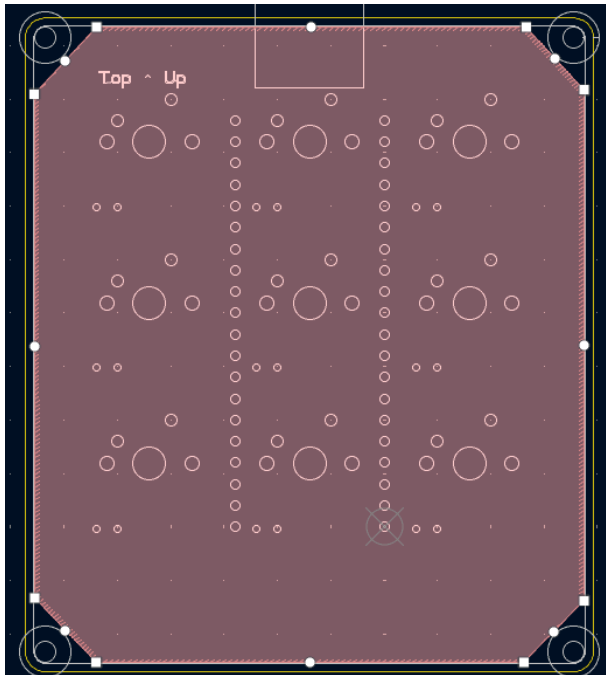


Figure 14: Front Filled Copper Zone

To create the remaining layer, **right click on an already completed plane** → **Zone** → **Duplicate Zone onto Layer** and then select the correct settings in the “Copper Zone Properties” window. Once that is complete you will have to **fill copper zones again by pressing the “B” key**.

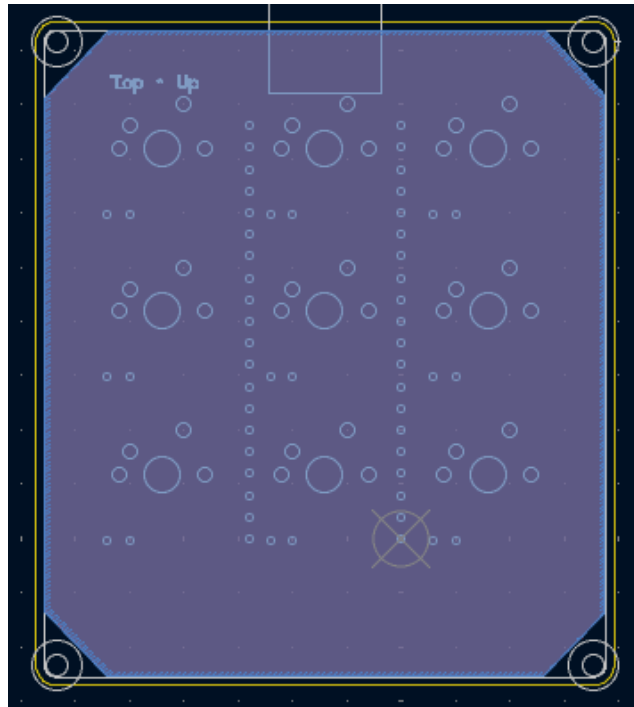


Figure 15: Back Filled Copper Zone

Double check that both layers have a copper outline by turning off one layer at a time from the “Appearance” panel on the right hand side.

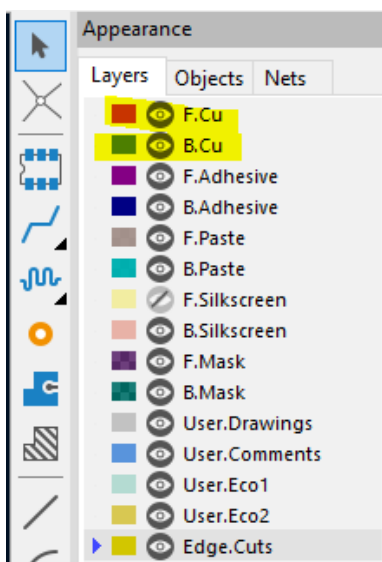


Figure 16: Appearance Panel

2.3 Aligning Components

In this section you will be learning how to place and align components on both sides of the board.

2.3.1 Placing and Aligning Components

Components can be placed anywhere on the board with respect to the mechanical outline and the copper boundaries. Since all components that are being used in the project are through hole components, you need to ensure that no two components overlap if one is placed on the bottom of the board and the other is placed on the top. The mechanical outline provides boundaries for where some components must be placed so that it can fit within the mechanical enclosure. These components are the drill holes for the screws, the switches and the microcontroller.

When placing components, it is easier if you start with the components that have mechanical constraints. In this project's case, that would be the switches and the microcontroller.

The Switches:

The switches are going to be placed on the top side of the board. Start by placing the grid origin in the middle of the switch, this helps with placing the component in the correct position to align properly with the mechanical outline. Then drag and drop the component and place it on top of the mechanical outline. **Place the grid origin by going to Place → Grid Origin.**

Note: You can toggle between turning on and off the ratsnest to have a clearer view of your board. A ratsnest are the unrouted wires connecting components before routing. To toggle ratsnest, go to the **Appearance tab on the right hand side → Objects → Ratsnets.**

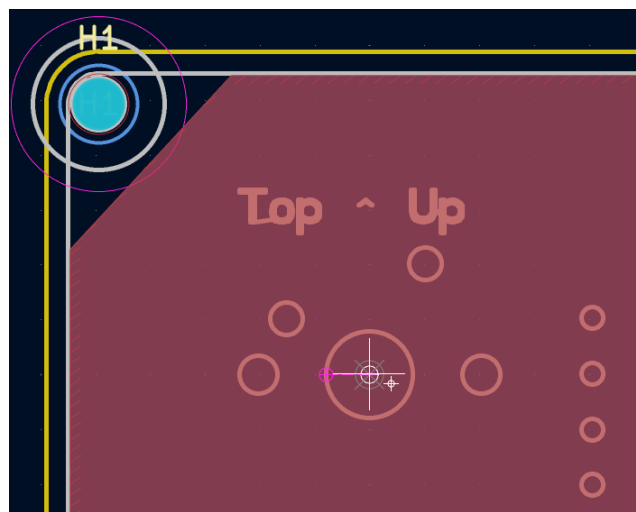


Figure 17: Grid Origin Placed in Switch

To make moving the component smoother, you can **click on the left Ctrl while you drag the component**. You can also change the grid size for more precise movement. Change the grid size by **right clicking anywhere on the sheet → Grid**.

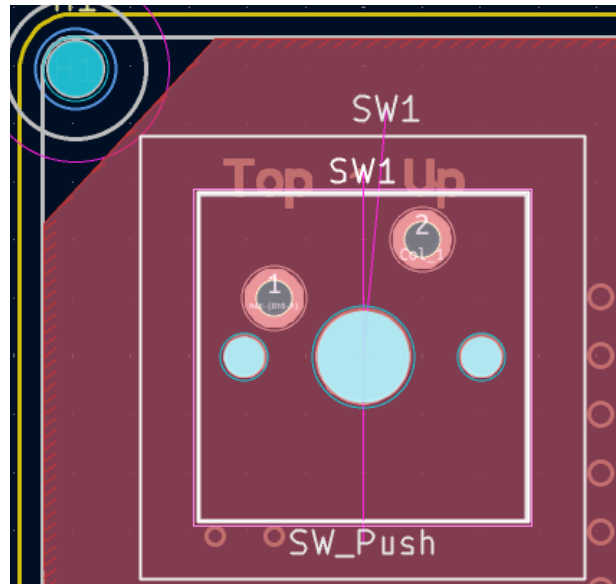


Figure 18: Switch Placed

Once the rest of the switches are placed, you can align the components to each other to ensure they are symmetrical. Vertical and horizontal components must be aligned separately. To align components, **select all the components in a row → right click on the component you want the rest to be aligned to → Align/Distribute → Align to Vertical Centre**. The same process is done for the vertical columns but the alignment option used will be different.

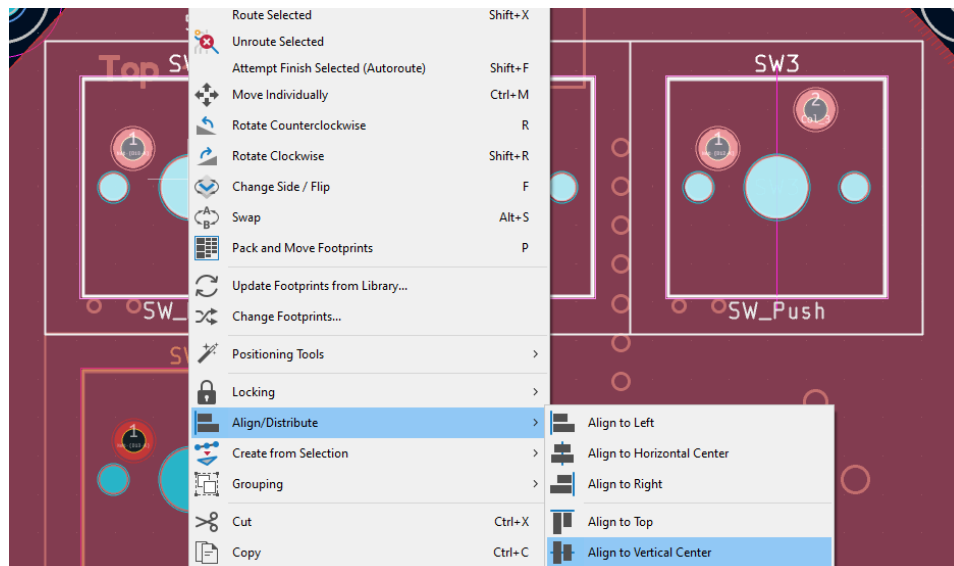


Figure 19: Aligning Switches

The Microcontroller:

The microcontroller will be placed on the bottom side of the board. To change the layer of a component, right click on the component → Change Side / Flip.

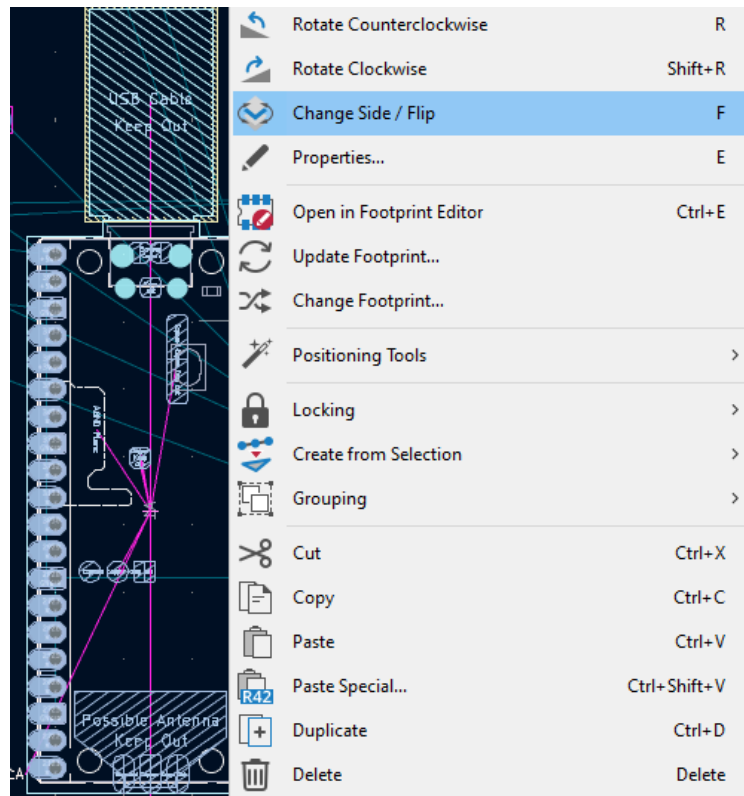


Figure 20: Changing Component Layer

Place the microcontroller on the mechanical outline using the same alignment steps as above.

Note: Once all the components are placed, **make sure to refill all copper zones by clicking “B”**. This will re-flow all the copper and take out the copper under any holes / traces.

2.3.2 Editing Footprints

For the purpose of this exercise, an example of editing the microcontroller footprint will be shown but note that the footprint *does not* need to be edited, it was already done in advance.

The microcontroller has multiple different keep-outs that will not be affecting the design and therefore need to be deleted. **Delete the “Possible Antenna” keep out boundary**, we will not be using an antenna for this design and it will provide more copper flow for board as well as extra space for tracing. We will also be **deleting the “Exposed Copper Keep Out” boundary**. This is because the microcontroller will not be touching the PCB, it will be lifted off using headers and therefore won’t affect the design.

To edit the footprint, **right click on the component** → **Properties** → **Edit Footprint** → **Select the keep out boundary and delete and save.**

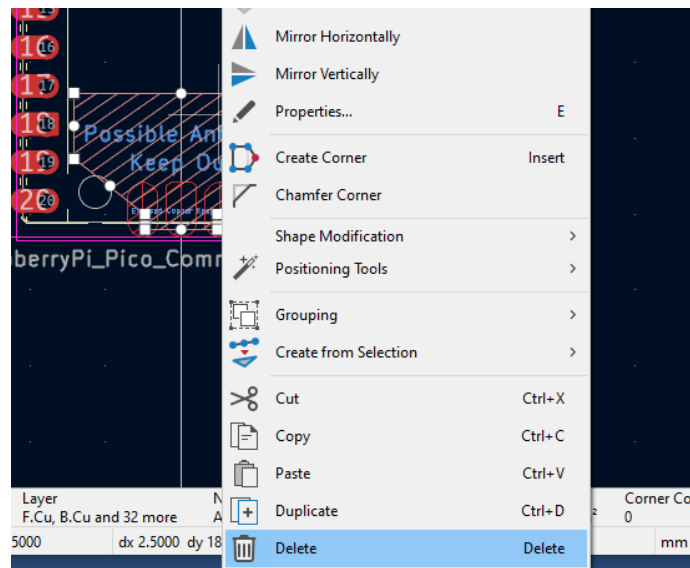


Figure 21: Editing Footprint

The microcontroller footprint after edits should look like the Figure 21 below.

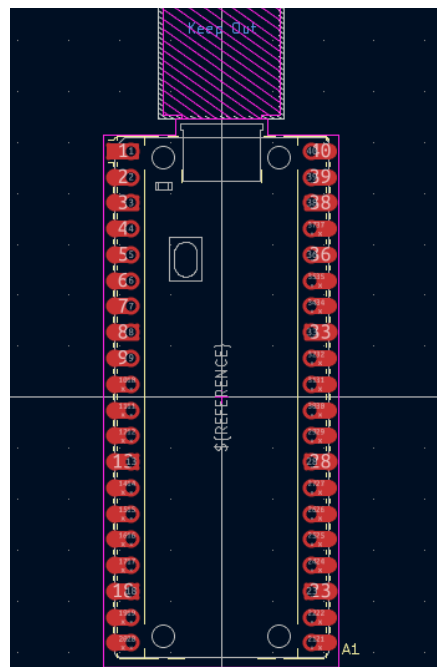


Figure 22: Final Microcontroller Footprint

2.3.3 Trace Width

Once all the components are placed, the next step is to connect the components together. The components are routed together using traces.

The trace width can be determined by the current the trace must carry, the temperature rise as well as the copper thickness. We also must keep in mind the minimum trace width that is set by the manufacturer.

For low-current signal traces, the typical trace width ranges from 0.007-0.012 inches. Power and ground traces need to be wider to safely carry higher current without overheating and causing damage. Wider traces also reduce voltage drop making sure the components get their required voltage levels. The power and ground trace widths typically range from 0.015 inches and up, larger currents require 0.030 inches and up.

This is a low-current project, therefore for the signal trace width we will be using **0.007874 inches (0.2mm)** and for the ground and power trace width we will use **0.0197 inches (0.5mm)**.

You can pre-set the trace width by going to File → Board Setup → Design Rules → Pre-defined sizes.

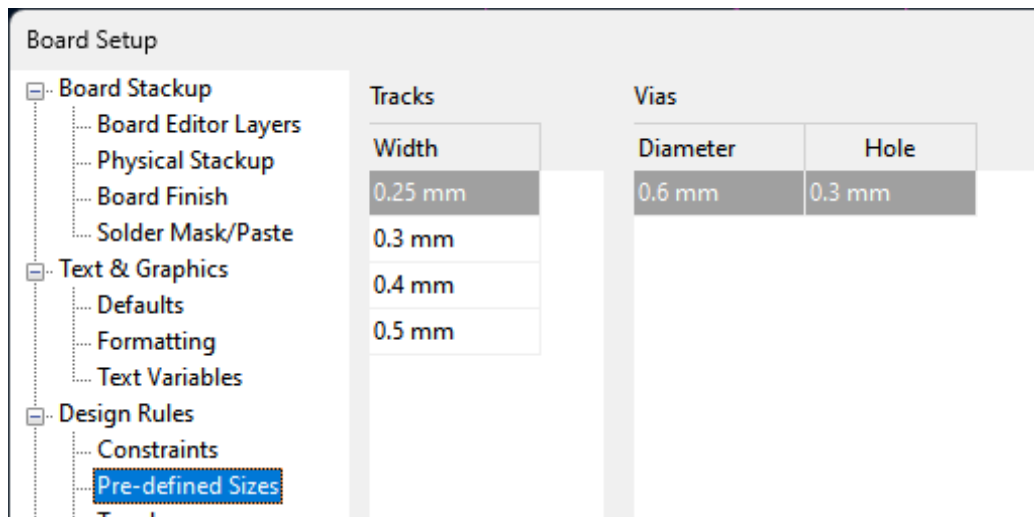


Figure 23: Defining Trace Width and Via Size

To place a trace go to Route → Route Single Track.

Switch the trace size using the “Track” drop down menu on the top panel.

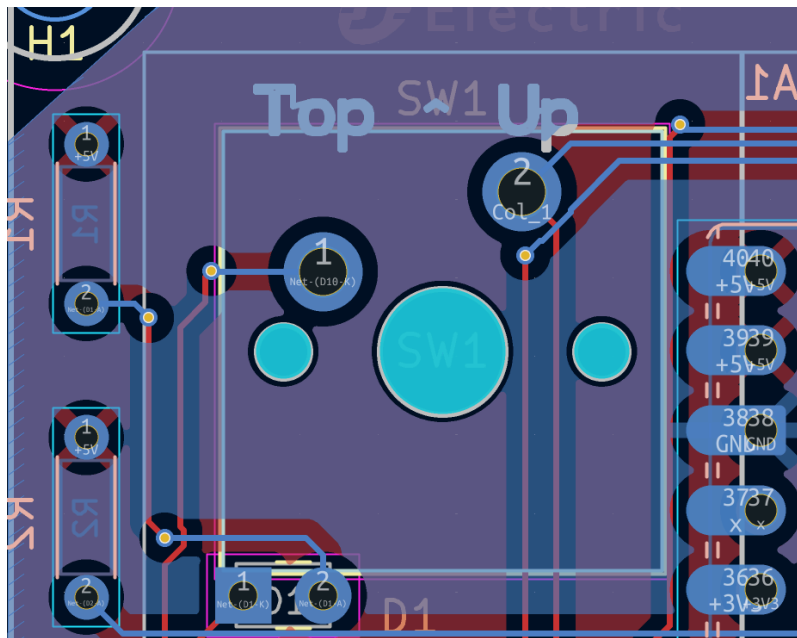


Figure 24: Example of Traces on the PCB

When placing a trace, you can toggle between layers as mentioned in the sections above. Typically with two layer boards, one layer would be dedicated to horizontal trace routes and the other would be for vertical routing.

2.3.2 Vias

Vias are small plated holes in a PCB that electrically connect different layers of the board together. This allows power and signals to pass vertically through the board. Traces connect to each other on different layers of the board through vias.

Vias are particularly useful for providing power distribution, space optimization, electrical connectivity, signal integrity (they minimize noise and signal distortion) and power distribution. The standard via size that will be used for this project is 0.6mm diameter and 0.3mm hole. Preset the via size **going to File → Board Setup → Design Rules → Pre-defined sizes.**

To continue a track from one layer of the board to another, **add a via by going to Place → Place Via.**

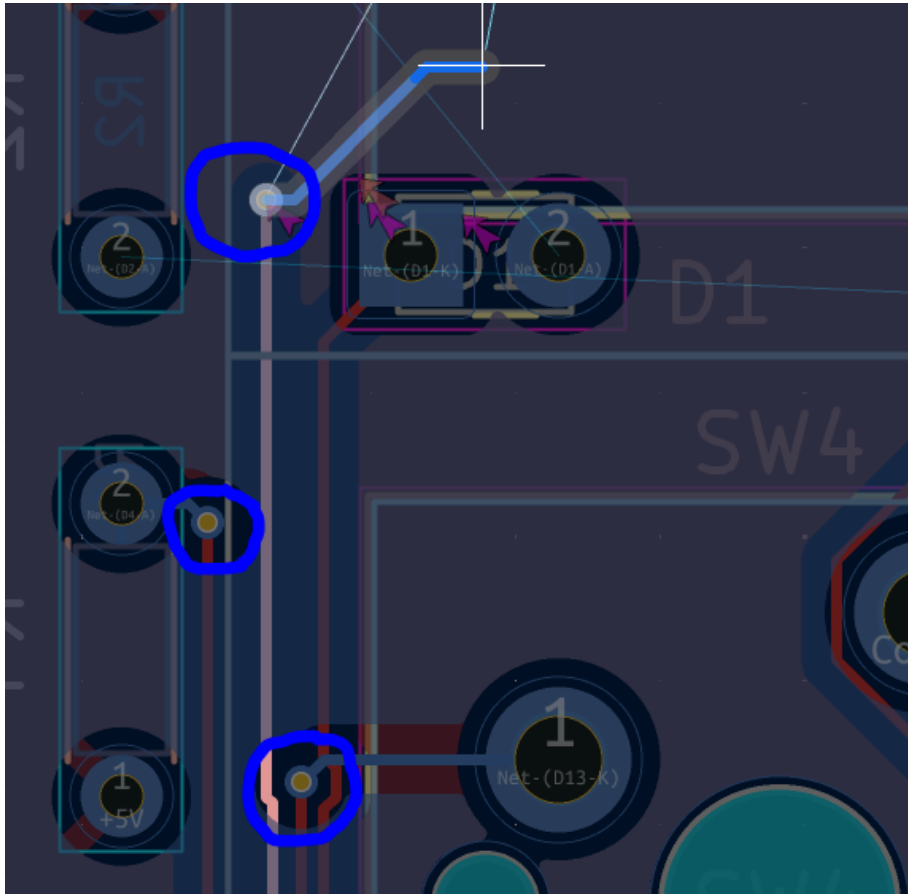


Figure 25: Example of Vias on the PCB

2.3.3 Copper Flow and Islands

Copper flow is one of the most important aspects of a layout design. The goal is to have a well-distributed and continuous copper plane to provide reliable signal transmission. An uneven copper flow could cause inaccuracies and signal degradation.

Copper islands are isolated areas of copper on the PCB that are not electrically connected to any net or trace. They unintentionally formed while routing and placing components that leave small, disconnected patches of copper. We need to make sure that they are not on the PCB by shifting routing / components to avoid them. Figure 25 is an example of a copper island and Figure 26 is an example of a good copper flow.

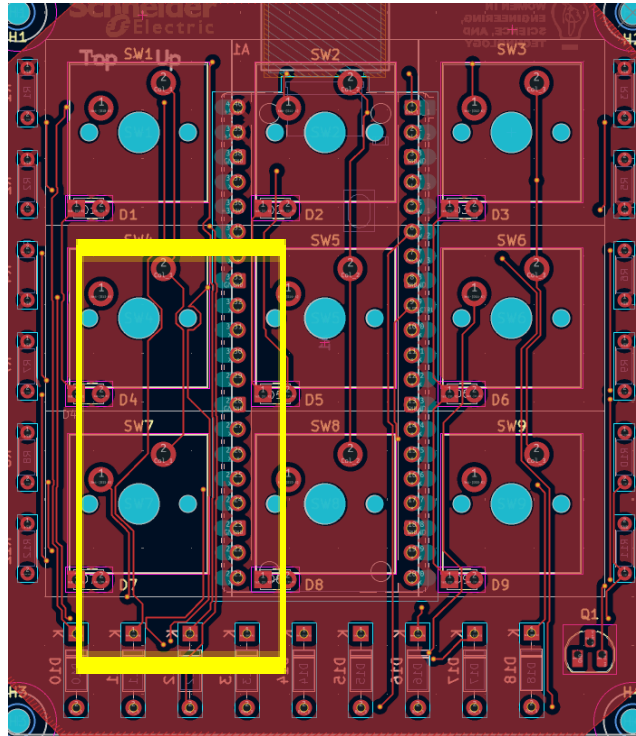


Figure 26: Example of a Copper Island

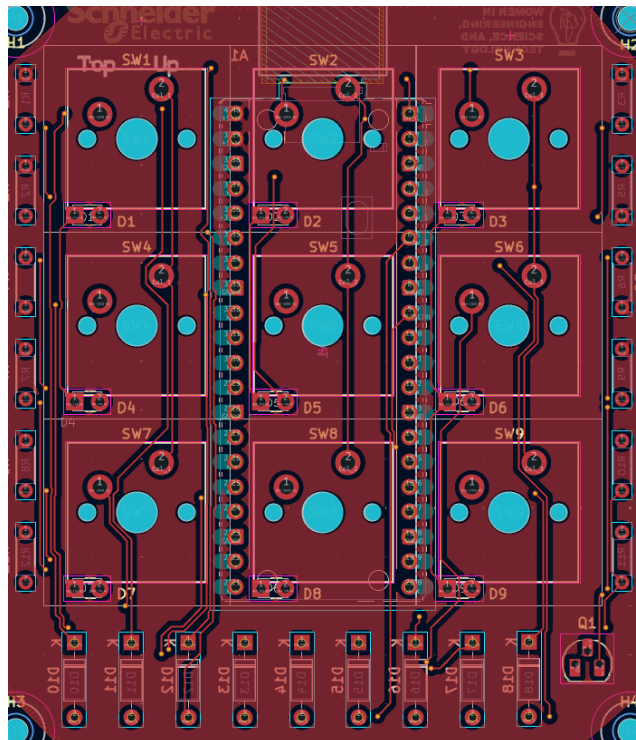


Figure 27: Example of Good Copper Flow

2.3.4 Design Rule Checker (DRC)

The design rule checker identifies any errors that are affecting the design. Common errors include components not connected, components boundaries are overlapping, components placed in a “keep out” zone etc. Some errors are expected and don’t need to be addressed as long as they are understood and justified. For example, in this design we intentionally placed the LEDs boundaries overlapping with the switch footprint to provide better backlight.

To check DRC’s go to Inspect → Design Rule Checker.

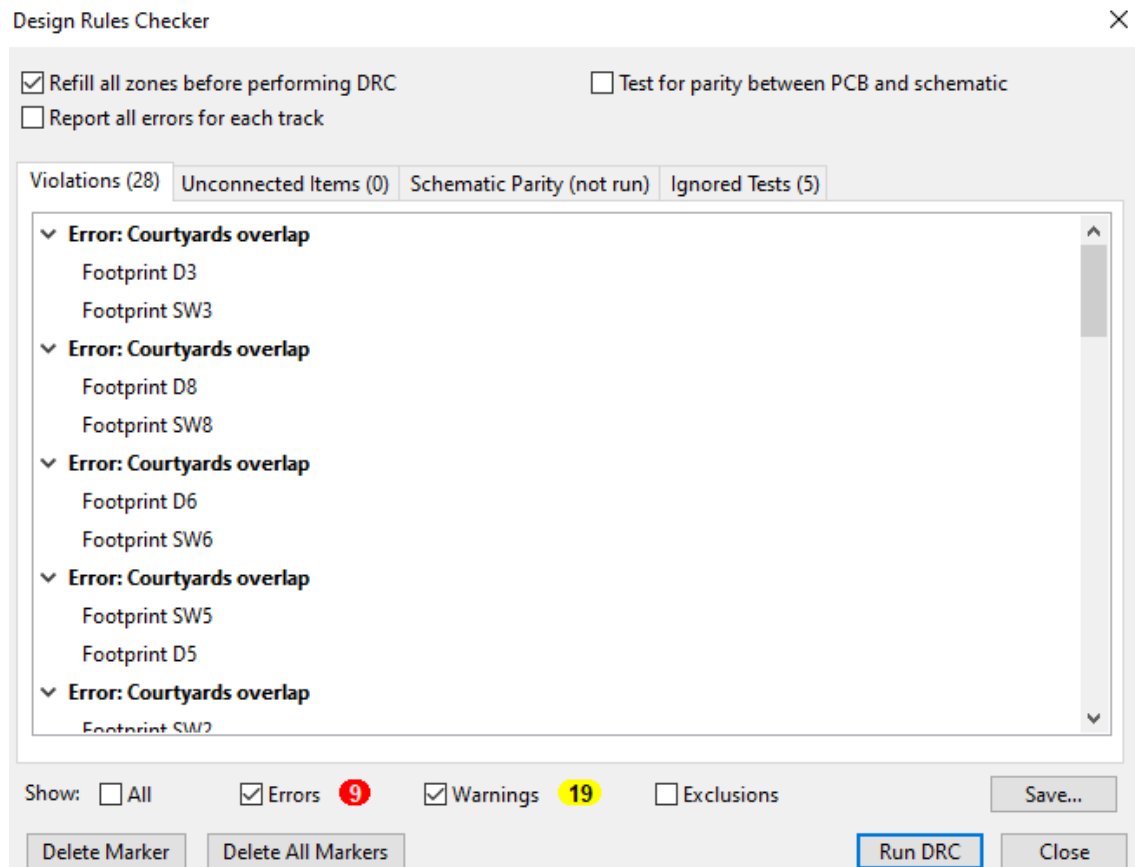


Figure 28: DRC Errors and Warnings

The Error “Courtyards overlap” and Warning “Silkscreen clipped by soldermask” that are referring to the switches and LEDs are intentional and will not affect the design.

2.4 Silk Screen

The silkscreen is the topmost layer of printed text, symbols, logos and graphics on the board’s surface. It’s used to indicate the components reference designators and polarity and it can

provide warnings or have special instructions as well as providing relevant information about the board.

2.4.1 Adding a Logo (optional)

To add a logo onto your design, go to the **main KiCad window** and click on **Image Converter** → **Load Source Image** → **Select Image** → **Output Format** → **Footprint** → **Edit size is desired** → **Export to File** – **Save**.

Go back to **layout design** and go to **Tools** → **Footprint Editor** → **File** → **Import** → **Footprint** → **Select Logo** → **Save into Footprint Library**.

Exit out of the Footprint Editor and go back to **layout design** and go to **Place** → **Place Footprints**, locate the logo and place it on the board. Ensure that it is on the Silkscreen layer.

Note: Make sure that the logo chosen is a transparent image and save as a PNG.

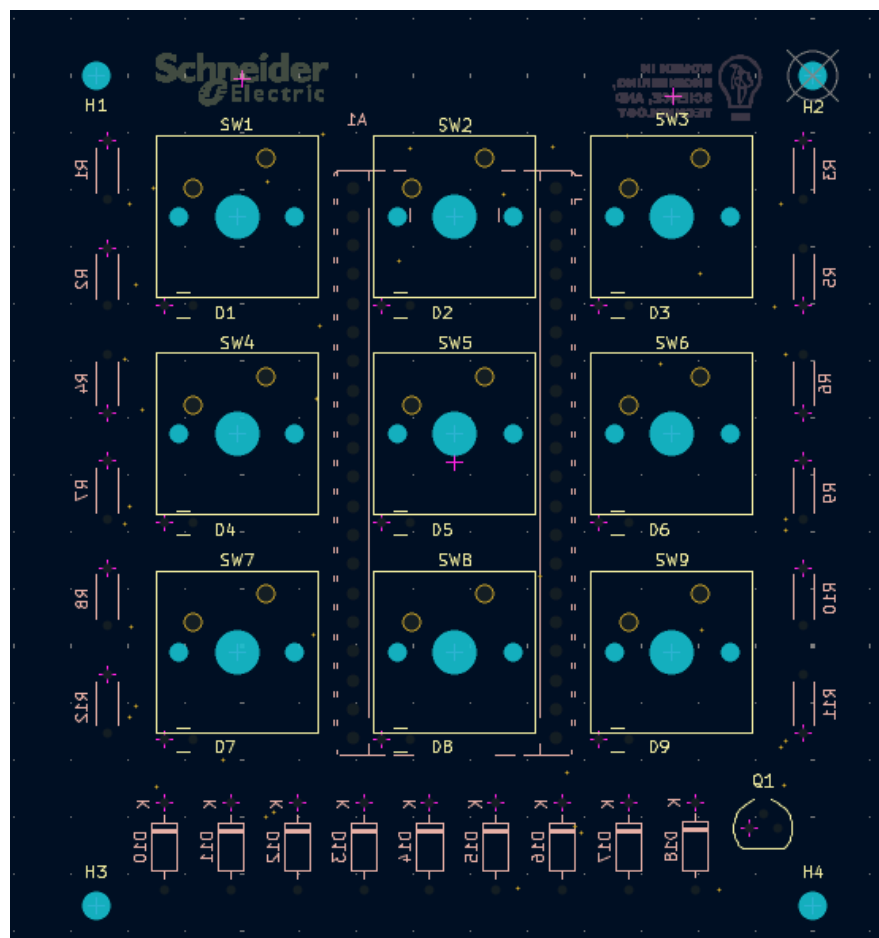


Figure 29: Top and Bottom Silkscreen Layers

3.0 Exporting Design

3.1 Drawing Sheet Table

The drawing sheet table is where you can edit the title block / information table on the bottom right side of the layout design. In that block, you can add what revision the board is in, the date, name as well as the designer. To edit the drawing sheet, **go to File → Page Settings**.

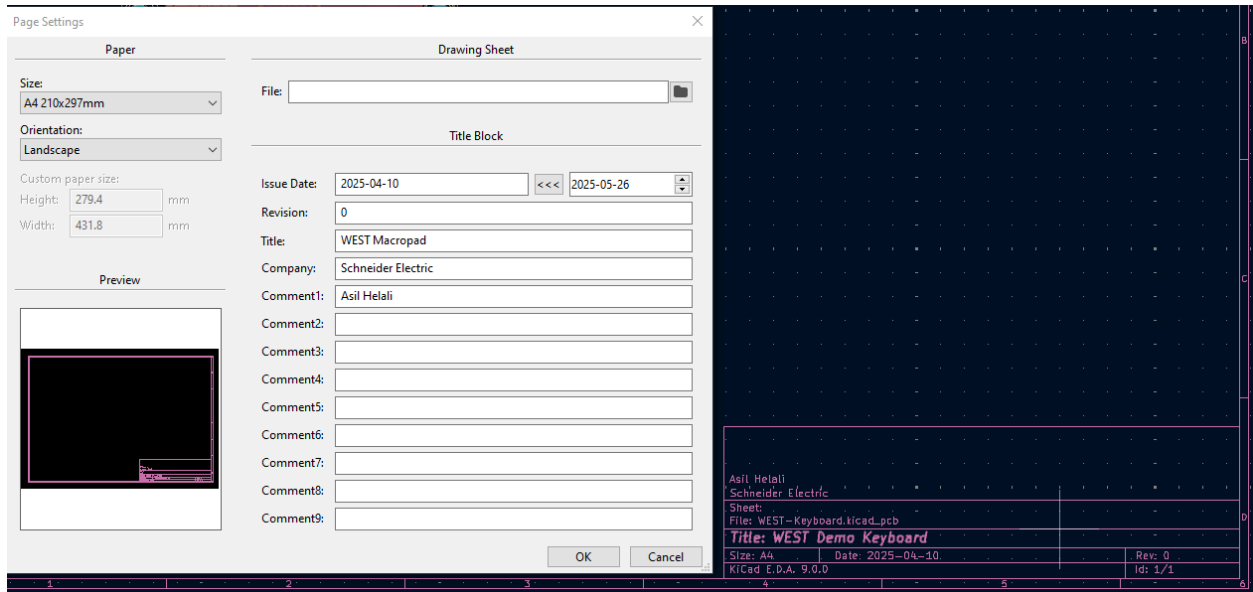


Figure 30: Page Information Settings

3.2 Gerber Files

Gerber files give you detailed information about each layer of a PCB. Each gerber file corresponds to one specific layer such as copper layers, solder paste, silkscreen, it also generates the drill hole information. It describes the exact sizes and shapes of all elements on all layers. These files are what each manufacturer uses when ordering a PCB board.

To export Gerber files go to File → Plot → Plot Format → Gerber → Chose an Output directory

Make sure the following layers are selected:

- F.Cu
- B.Cu
- F.Paste
- B.Paste

- F.Silkscreen
- B.Silkscreen
- F.Mask
- B.Mask
- Edge.Cuts

Click on “Generate Drill Files” once that is complete, click on “Plot”

Double check the gerber files by going to the **KiCad window → Gerber Viewer → File → Open Gerber Plot Files** and selecting the files to view.

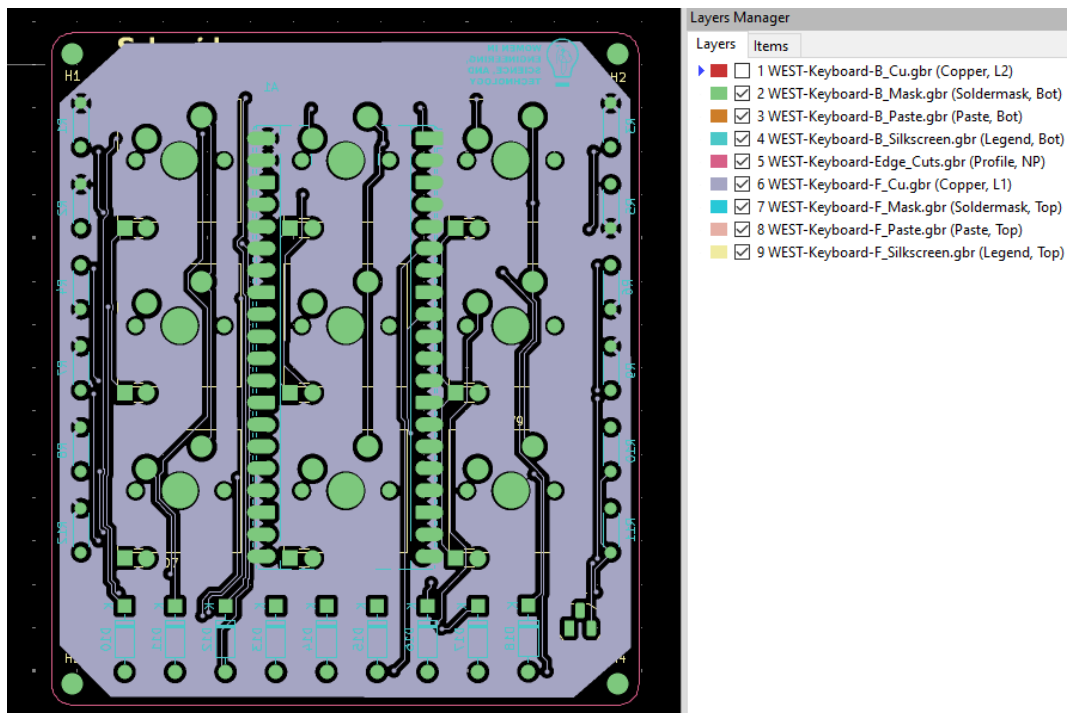


Figure 31: Gerber Viewer

3.3 Exporting Mechanical STEP File (optional)

.STEP (Standard for the Exchange of Product Data) are files that contain the 3D model of the PCB. They are mostly used for mechanical design purposes. This file can be uploaded to Fusion 360 and the enclosure can be built around it.

To export a .STEP file go to File → Export → STEP.

Note: To view the 3D model, go to **PCB Editor Window → View → 3D Viewer.**

3.4 Final Results

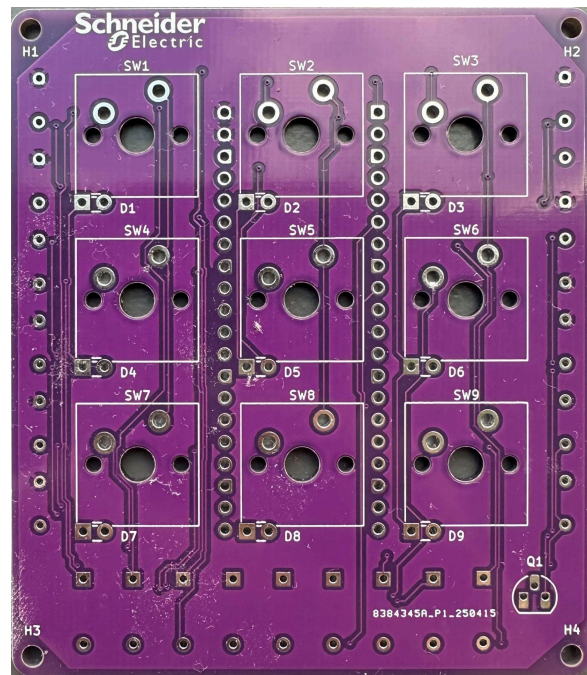


Figure 32: Final PCB Front Side

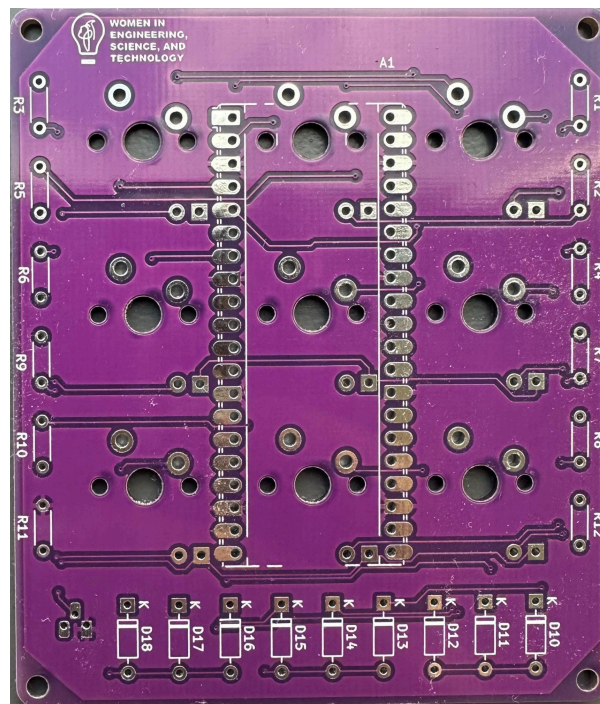


Figure 33: Final PCB Back Side

4.0 Check-List

- ☐ Annotate the schematic
- ☐ Board constraints are set
- ☐ Mechanical outline is uploaded
- ☐ Footprint library is updated (not needed)
- ☐ Board outline
- ☐ Ground copper plane
- ☐ Power (+5V) copper plane
- ☐ Are the copper planes at least 0.5mm away from the board outline?
- ☐ Copper plane is NOT overlapping with the screw holes
- ☐ Components are on the board
- ☐ Components are exactly aligned with the mechanical outline
- ☐ Are all components routed?
- ☐ Do the power and ground pins on components flow nicely on the plane?
- ☐ Are there any copper islands?
- ☐ Is there good copper flow?
- ☐ No unexpected errors when running the DRC
 - ☐ If there are DRC's, are they justified?
- ☐ Are all reference designators clearly placed on the silkscreen
- ☐ Did you add a logo? (optional)
- ☐ Gerber files exported
- ☐ .STEP file exported (optional)

5.0 Resources

- KiCad 9.0 Layout Reference Manual -
<https://docs.kicad.org/9.0/en/pcbnew/pcbnew.html>
- “An Intro to KiCad” Tutorial videos , (parts 4 - 8 are relevant to layout design) -
<https://www.youtube.com/watch?v=4Gtd7xY6zS4&list=PL3bNyZYHcRSUhUXUt51W6nKvxx2ORvUQB&index=3>
- PCB Stack-Up Information -
<https://www.sfcircuits.com/pcb-production-capabilities/pcb-stack-up>
- PCB Manufacturer that will be used for this project (JLCPCB) -
https://jlcpcb.com/?from=VGB&utm_source=google&utm_medium=cpc&utm_campaign=13448786080&utm_content=704829674800&utm_term=b_jlcpcb&adgroupid=120688669102&utm_network=g_&gad_source=1&gad_campaignid=13448786080&gbraid=0AAAAACUKBgQEkQe7n7wWmOcQ0IJNw_PSO&gclid=EAIaIQobChMIhaOLjMzEjQMVm9HCBB1r1xiWEAAYASAAEgIwR_D_BwE